

DATABASE MANAGEMENT SYSTEM

Database Management Systems: Relational & Distributed Data

Relational Algebra, SQL Optimization, Normalization (1NF to BCNF), ACID Transactions & Indexing

Academic Publication Details:

Subject: Database Management System

Principal Author / Academic Reviewer: Prof. Rajeshwari Das & Database Systems Group

Standard: Quality Education Curriculum Standard

Format: Official Open Access Educational Material

Table of Contents

Chapter 1: Relational Model, Relational Algebra & Advanced SQL

Core Competency: Relational Foundations

Chapter 2: Database Design, Functional Dependencies & Normalization

Core Competency: Schema Design & Normalization

Chapter 3: Transactions, ACID Properties, Concurrency & Indexing

Core Competency: Transaction Processing & Indexing

CHAPTER 1

Relational Model, Relational Algebra & Advanced SQL

1. Theoretical Foundations & Scope

Relational database systems form the foundation of enterprise data management. This chapter covers the formal relational model, relational algebra operators (select, project, join, division), and advanced SQL querying techniques.

2. Core Conceptual Framework

- **Relational Algebra Operators:** Formal mathematical operations underpinning SQL query parsing and execution planning.
- **Integrity Constraints:** Primary keys, foreign keys, unique constraints, and referential integrity cascade actions.
- **Complex SQL Queries:** Correlated subqueries, window functions (ROW_NUMBER, RANK), and common table expressions (CTEs).

Key Practical Insights & Examination Takeaways:

- ' Translate declarative SQL queries into procedural relational algebra expressions.
- ' Enforce strict entity and referential integrity constraints across schema tables.
- ' Write optimized multi-table JOINS and analytical aggregation queries.

CHAPTER 2

Database Design, Functional Dependencies & Normalization

1. Theoretical Foundations & Scope

Good schema design minimizes data redundancy and avoids insertion, deletion, and update anomalies. We examine functional dependencies, Armstrong axioms, attribute closures, and normalization forms: 1NF, 2NF, 3NF, and Boyce-Codd (BCNF).

2. Core Conceptual Framework

- **Database Anomalies:** Redundancy-induced update, insertion, and deletion anomalies in unnormalized tables.
- **Functional Dependencies:** Determining candidate keys by computing attribute closures with Armstrong axioms.
- **1NF, 2NF, 3NF & BCNF:** Progressively eliminating repeating groups, partial dependencies, and transitive dependencies.

Key Practical Insights & Examination Takeaways:

- ' Decompose unnormalized relations into lossless, dependency-preserving BCNF schemas.
- ' Compute minimal covers and candidate keys from sets of functional dependencies.
- ' Balance normalization purity with denormalization performance considerations.

CHAPTER 3

Transactions, ACID Properties, Concurrency & Indexing

1. Theoretical Foundations & Scope

Enterprise databases require concurrent multi-user access while maintaining correctness. This chapter covers ACID properties, serializability, Two-Phase Locking (2PL), deadlock handling, write-ahead logging (WAL), and B+ tree indexing.

2. Core Conceptual Framework

- **ACID Guarantees:** Atomicity, Consistency, Isolation, and Durability guarantees enforced by storage engines.
- **Conflict Serializability & 2PL:** Testing conflict serializability via precedence graphs and enforcing 2-Phase Locking.
- **B+ Tree Indexing:** Balanced search tree structures storing all keys in linked leaf nodes for rapid range queries.

Key Practical Insights & Examination Takeaways:

- ' Construct serialization graphs to verify whether concurrent schedules are conflict serializable.
- ' Explain how Write-Ahead Logging (WAL) ensures durability across unexpected system crashes.
- ' Analyze query execution plans to identify when B+ tree index scans outperform full table scans.

Academic References & Further Study

[1] Quality Education Academic Standards & Pedagogical Guidelines, 2026 Edition.

[2] International Computer Science & Engineering Curriculum Framework.

[3] Educational Technology & Assessment Methodology Research Series.

[4] Open Access Educational Resource Repository - Quality Education Platform.

